EX. NO: 5

IMPLEMENTATION OF PLAYFAIR CIPHER

AIM:

To write a C program to implement the Playfair Substitution Cipher technique.

ALGORITHM:

1. Start the program.
2. Enter the keyword and plaintext.
3. Create a 5 × 5 matrix using the keyword.
4. Fill the remaining spaces with unused letters of the alphabet, combining `I` and `J`.
5. Divide the plaintext into pairs of letters.
6. If both letters in a pair are the same, insert `X` between them.
7. If the letters are in the same row, replace them with the letters on their right.
8. If the letters are in the same column, replace them with the letters below them.
9. Otherwise, replace each letter with the letter at the opposite corner of the rectangle.
10. Display the encrypted text.
11. Stop the program.

PROGRAM:

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>



char matrix[5][5];



void main()
```

```
{
    char key[100], plain[100];
    int used[26] = {0};
    int i, j, k = 0;
    int r1, c1, r2, c2;
    char ch, a, b;

    printf("Enter the key: ");
    scanf("%s", key);

    printf("Enter the plain text: ");
    scanf("%s", plain);

    /* Create 5 x 5 Playfair Matrix */

    for(i = 0; key[i] != '\0'; i++)
    {
        ch = toupper(key[i]);

        if(ch == 'J')
```

```
        ch = 'I';

    if(ch >= 'A' && ch <= 'Z' && !used[ch - 'A'])
    {
        matrix[k / 5][k % 5] = ch;
        used[ch - 'A'] = 1;
        k++;
    }
}

for(ch = 'A'; ch <= 'Z'; ch++)
{
    if(ch == 'J')
        continue;

    if(!used[ch - 'A'])
    {
        matrix[k / 5][k % 5] = ch;
        used[ch - 'A'] = 1;
        k++;
```

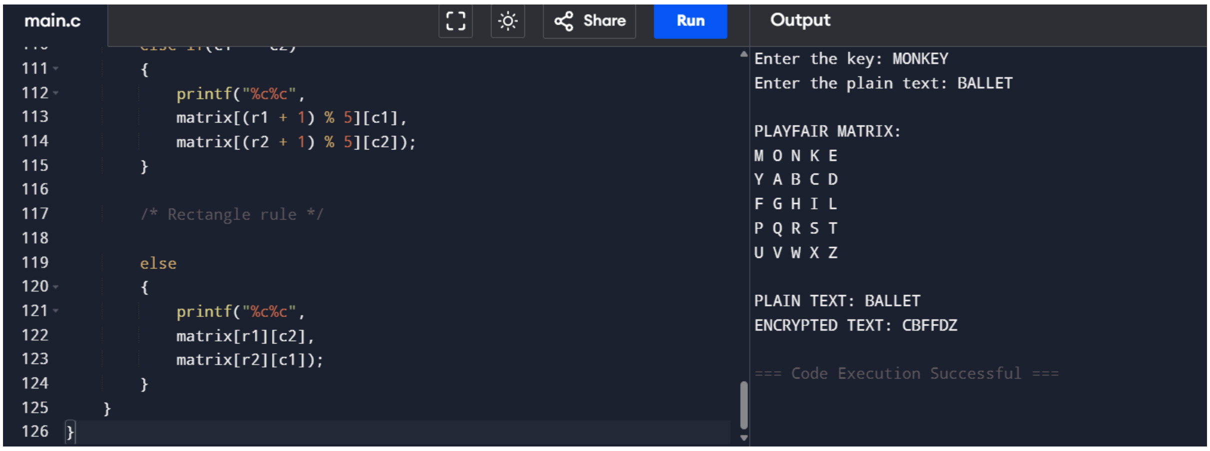
```
    }
}

printf("\nPLAYFAIR MATRIX:\n");

for(i = 0; i < 5; i++)
{
    for(j = 0; j < 5; j++)
        printf("%c ", matrix[i][j]);

    printf("\n");
}

printf("\nPLAIN TEXT: %s", plain);
printf("\nENCRYPTED TEXT: ");

for(i = 0; plain[i] != '\0'; i += 2)
{
    a = toupper(plain[i]);
```

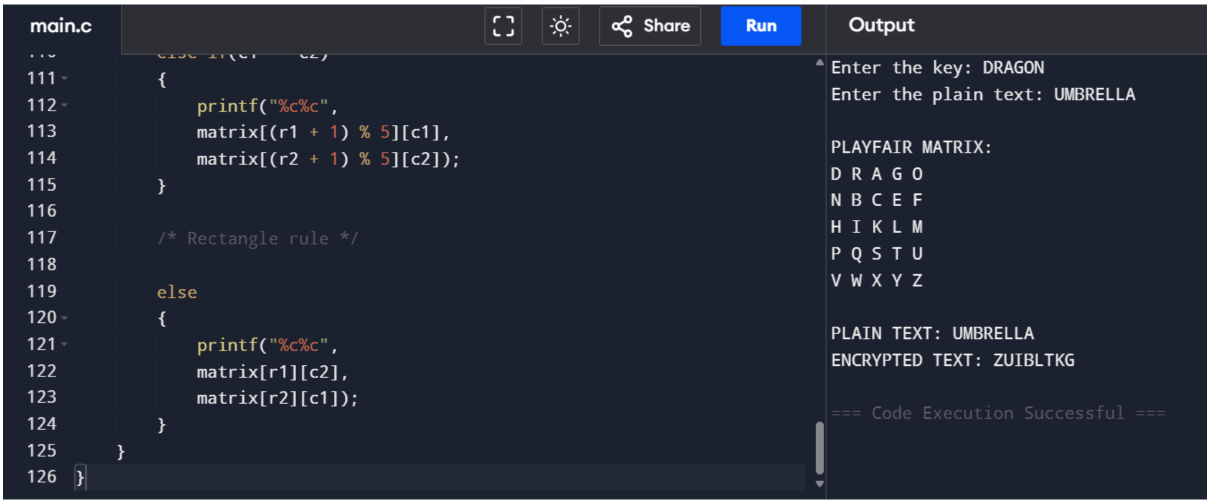
```
if(plain[i + 1] != '\0')
    b = toupper(plain[i + 1]);
else
    b = 'X';


if(a == 'J')
    a = 'I';


if(b == 'J')
    b = 'I';


/* Find positions of the two letters */


for(j = 0; j < 5; j++)
{
    for(k = 0; k < 5; k++)
    {
        if(matrix[j][k] == a)
        {
            r1 = j;
```

```
            c1 = k;
        }

        if(matrix[j][k] == b)
        {
            r2 = j;
            c2 = k;
        }
    }
}

/* Same row */

if(r1 == r2)
{
    printf("%c%c",
    matrix[r1][(c1 + 1) % 5],
    matrix[r2][(c2 + 1) % 5]);
}
```

```
        /* Same column */

        else if(c1 == c2)
        {
            printf("%c%c",
            matrix[(r1 + 1) % 5][c1],
            matrix[(r2 + 1) % 5][c2]);
        }

        /* Rectangle rule */

        else
        {
            printf("%c%c",
            matrix[r1][c2],
            matrix[r2][c1]);
        }
    }
}
```

OUTPUT:





RESULT:

Thus, the Playfair Substitution Cipher technique was successfully implemented using C language, and the given plaintext was successfully encrypted using the generated 5 × 5 Playfair matrix.